EQUIPMENT NAME:	**FIM III Booster Station Transformer**
EQUIPMENT NUMBER:	**F23-TR-602**

Procurement Package		"EL006" – 11/0.433 kV Dry Type Transformer		
Equipment No.	**Rating (kVA)**	**Voltage Ratio (kV)**	**Vector Group**	**Equipment Name**
F23-TR-602	2500	11/0.433	Dyn11	FIM III Booster Station 11/0.433 kV Transformer
Reference Documents		2000-F23-SLD-EL-40011 – FIM III Booster Station LV Motor Control Centre Single Line Diagram Sheet 1 of 2		
Reference Specifications		2000-F00-STS-EL-10017 – Dry Type Power Transformer Specification 2000-F00-STS-GE-10002 - Site Specific Information 2000-F00-STS-GE-10001 - Packing and Shipping Specification SPEC-00-G-05 – Protective Coating		

Scope of Supply

Work Included

- Design, supply, and fabrication of One (1) Step down dry type transformer including, but not limited to:
  - Transformer IP2x fully assembled complete with:
  - Tap changer (links)
  - Cooling fans (ANAF)
  - Cable clamps (as required for bottom entry)
  - Earth studs for core and frame
  - Appropriately sized Earth bar located on HV side. Allowance to be made for all required Earths (Incoming cables Sheath/screen, Surge Arrestors, Transformer Earthing, Switchroom Earth bar link)
  - Polymer/silicone surge arrestors for HV winding surge protection – pre-installed on appropriate brackets
  - Mounting lugs for affixing to switchroom structural steel
  - Instrumentation (winding temperature) and junction boxes
  - Wiring for instrumentation and controls including fan starters if required to comply with the environmental specification
  - Labelling and nameplate displaying all transformer data
  - Lifting lugs with Rated Capacity label ("Rated Capacity" definition as per 1418.8-2008)
  - Any other items which are normally supplied with this equipment
- Manufacturer's standard Surface Protection equivalent to or better than Coating System CS1 in SPEC-00-G-05 – Protective Coating specification
- All ancillary items and special tools normally supplied with, or required to adjust and maintain all components associated with the equipment
- Laser engraved 316SS Stainless steel equipment nameplate
- All fasteners, gaskets, packers, shims and any other items necessary for site assembly
- Factory Acceptance Testing at manufacturer's premises
- Supply of all necessary documentation, drawings, and test documentation
- Packaging suitable for sea and road transport for delivery
- Spare parts list
- Delivery to the delivery point.

Work Excluded

- Hold-down bolts, nuts, and washers at concrete foundations/structural steel
- Offboard cabling from Transformer to switchroom
- Offloading the equipment at vendor premises
- Installation and commissioning at site

	Unit	Duty	Vendor	Notes	Rev
General					
Transformer Electrical Ratings	kVA	1600		1	0
Transformer Type		Resin Encapsulated, dry-type	Cast resin dry type	1	0
Equipment Number and Name Plate		Yes, on stainless steel plate	Yes, aluminium alloy	1	0
Make		Siemens	Siemens Energy	3	0
Model		Cast Resin Dry Type	Cast resin dry type	3	0
Country of Manufacture			China	2	0
Compliance with Australian Standards	Comply / Do Not Comply	Yes, AS2067, AS2374, AS60076, AS60214, AS60137 vendor to state compliance	Yes, AS2374, AS60076, IEC60076	1	0
Environmental Conditions					
Plant Location		Western Australia	western Australia	1	0
Service Location	Indoor / Outdoor	Indoor, within a switchroom in a temperature controlled, air-conditioned environment	Comply	1	0

	Unit	Duty	Vendor	Notes	Rev
Site Data		Refer to 2000-F00-STS-GE-10002 – Site Specific Information	Comply	1	0
Physical Construction (Fluid Filled)					
Core and Frame Material		Corrosion Resistant Steel	Corrosion resistant steel	3	0
Mounting		Within switchroom on steel structure		1	0
Cooling		Air Natural/Air Forced (ANAF)	AN/AF	3	0
IP Rating of Enclosure	IPXX	IP00 (Indoor)	IP00 (indoor)	3	0
Lifting Lugs	Yes / No	Yes	Yes	3	0
Jacking Lugs	Yes / No	Yes	Yes	3	0
Frame Mounted Wheels	Yes / No	Yes	Yes	3	0
Transformer Painting Specification		As per manufacturer's standard	As per manufacturer's standard	3	0
Transformer Paint Colour (Enclosure)		RAL 7035	No enclosure	3	0
HV Winding Conductor		Aluminium		3	0
LV Winding Conductor		Aluminium	Aluminium	3	0
Temperature Rating					
Rated Maximum Ambient Temperature	ºC	Up to 50 in extreme circumstances when aircons are off/tripped – vendor to advise limitations	Comply	3	0
Winding Temperature Rise	ºC	90	90	3	0
Ability to Withstand Short Circuit in accordance with IEC 60076-5					
Ability to withstand dynamic effects resulting from a primary or secondary short circuit	Yes / No	Yes	Yes	1	0
Ability to withstand thermal effects resulting from a primary or secondary short circuit	Yes / No	Yes	Yes	1	0
Electrical Supply					
Primary Voltage	kV	11 ± 10%, 3 Phase	11+/-10%, 3 phase	1	0
System Earthing, Primary Side		11kV Impedance earthed to limit current to 25 Amps	comply	1	0
Secondary Voltage (fully loaded)	kV	0.433 ± 10%, 3 Phase, 4 wire	0.433+/-10%, 3 phase, 4 wire	1	0
System Earthing, Secondary Side		Solidly Earthed	Solid earthed	1	0
Load Characteristic		Active Harmonic Filtered	Active harmonic filtered	1	0
Full Load Rating	kVA	1600	1600	1	0
Vector Group		Dyn11	Dyn11	1	0
Impedance	%	To AS/NZS 60076.5	6% as per IEC60076.5	3	0
Frequency	Hz	50 ± 2%	50+/-2%	1	0
HV Winding Insulation					
Nominal	kV rms	11	11	3	0
Ratings	kV rms	12	12	3	0
Lightning-impulse withstand voltage (BIL)	kV peak	95	95	3	0
Power-frequency withstand voltage	kV rms	28	28	3	0
LV Winding Insulation					
Ratings	kV rms	1	1	3	0
Lightning-impulse withstand voltage (BIL)	kV peak	As per AS/NZS 61439.1	N/A for low voltage side	3	0
Power-frequency withstand voltage	kV rms	2	3	3	0
Tap Changer					
Type		Off-load, Manual, provided with handled operable from outside tank	off load tap link, no tank for dry transformer	1	0
Tapping Range	%	-5 to +5	-5 to +5	3	0
Step size	%	2.5	2.5	3	0
Location		HV Winding	HV winding	1	0
LV Winding Star Point Neutral CT		No	No	1	0
Fire, Environmental, Climate and Fire Behaviour Class					
Environmental Class		E2	E2	3	0
Climate Category		C2	C2	3	0
Fire Behaviour Class		F1	F1	3	0
Cable Connections					
Termination Details		Tinned flat copper bar, bolted cable lugs	tinned flat copper bar, bolted cable lugs	3	0
HV Cable Connections					
Number of cables		1	1	1	0

	Unit	Duty	Vendor	Notes	Rev
Insulation voltage	kV	6.35/11	Noted	1	0
Type of cable		3C, XLPE	Noted, cable is excluded from manufacture scope	1	0
Size of Conductor	mm^2	120	Noted, cable is excluded from manufacture scope	1	0
LV Cable Connections					
Number of cables		Up to 6 x 1C per phase/neutral	Noted, cable is included In switchroom scope	1	0
Insulation voltage	kV	0.6/1	Noted, cable is included In switchroom scope	1	0
Type of cable		X-110 Flex single cores	Noted, cable is included In switchroom scope	1	0
Size of Conductor	mm^2	Up to 500	Noted, cable is included In switchroom scope	1	0
Equipment and Fittings					
Thermal Protection	Yes / No	Yes, with Temperature Controller	Yes, with temperature controller	3	0
Winding Temperature Indicator – 6 x PT100 RTDs (2 per phase) with Thermal Relay and Alarm and Trip Contacts	Yes / No	Yes, with Temperature Controller	Yes, with temperature controller	3	0
Thermal Protection Relay Make and Model		PT100/NT935	Pt100/NT935	3	0
Forced Air Cooling Fans (Thermal Operation)		Controlled by Temperature Controller	Controlled by temperature controller	3	0
Cooling Fan Supply Voltage	VAC	240	240	3	0
Earthing Terminal on Core	Yes / No	Yes	Yes	3	0
Rating and Connection Diagram Plate	Yes / No	Yes, Aluminium Alloy	Yes, Aluminium alloy	3	0
Control Circuit:					
Marshalling Box / Control Panel		Yes, any instruments must be terminated in a junction box	Yes, with junction box	3	0
Trip and Alarm Voltage (protection DC voltage)	V	24VDC	24VDC	1	0
Instrumentation Connections		Screw Cage Terminals WDU2.5 or equivalent	As per manufacture standard	3	0
Enclosure/Accessories					
Enclosure Material		No Enclosure – Transformer will be housed in a sealed air-conditioned room	No enclosure	3	0
Cable Entry Location	Top / Bottom	Bottom	Bottom	3	0
HV Surge Arrestors	Required	Required on HV terminals- Vendor to provide surge arrestor details/specs	Yes	3	0
Test Requirements:					
Factory Acceptance Test		Yes, test certificates required	Not included	1	0
Type Tests per AS/NZS 60076		Yes, test certificates required	Not included	1, 4	0
Standard Routine Tests per AS/NZS 60076		Yes, documentation required	Yes	1	0
Shipping and Delivery Information					
Dimensions	mm	Length x Width x Height	2250*1087*1818	2	0
Weight	kg		Approx. 3670	2	0

Notes
1. Fixed. The Vendor shall comply with this data, as a minimum. Vendor shall state their applicable ratings/data or compliance.
2. Vendor to Advise.
3. Preferred, Vendor to advise and provide reasoning for difference.
4. Type test certificates to be provided with tender documentation

Vendor Drawing and Document Requirements List (VDDR)

E – Electronic Copy H – Hard Copy MDR – Manufacture's Data Report

FAT – Factory Acceptance Testing IOM – Installation and Operating Manual

VDDR			Required with Tender	Required after Award		As Built Required
			Copies	Copies	Weeks	Copies
A		**Production Schedule**				
	A01	Production Schedule / Bar Chart		1E	2	
	A02	Monthly Report		Monthly		
B		**Registers and General Drawing Requirements**				
	B01	Drawing Schedule / Register (Drawings Expected)		1E	2	
	B02	General Arrangement with Foundation Loads (in AutoCAD 2010 and PDF)		1E	6	1E
	B03	Cross-section		1E	6	1E
C		**Mounting and Erection Drawings**				
	C01	Foundation and Mounting Details (Provided in B02)		1E	6	1E
	C02	Pipe Spools				
	C03	Outline Dimensions, Cut-out and Support Details (Provided in B02)		1E	6	1E
	C04	Match Marked Erection Drawings				
	C05	Motor GA (with Fixing and Shaft Details, and Overall Dimensions)				
D		**Electrical Drawings (in electronic form as appropriate)**				
	D01	Single Line Diagrams				
	D02	Schematic Drawings				
	D03	Terminations / Loop Diagrams		1E	6	1E
	D04	Equipment Layout and Fabrication Details		1E	6	1E
	D05	Nameplate		1E	6	1E
	D06	Cable Block Drawings				
	D07	Electrical and Instrumentation GAs				
E		**Process and Piping Drawings**				
	E01	Process Flow Sheets				
	E02	Process and Instrument Diagram				
	E03	Pneumatic or Hydraulic Diagram				
	E04	Lubrication Diagram				
F		**Technical Data**				
	F01	Equipment Information Schedule	1E			
	F02	Electrical Performance data sheet		1E	6	
	F03	Equipment List		1E	6	1E
	F04	Vendor's Surface Protection Details	1E	1E	6	
	F05	Lubrication Data Schedule				
	F06	Material Schedule				
	F07	Bolt Schedule				
	F08	Fan Performance Curves				
	F09	Pump Performance Curves (with Duty Points Marked)				
	F10	Cable Schedule				
	F11	Vendor Catalogue Information	1E			
	F12	Calculations				
	F13	Completed $_{\text{Engineer}}$ Datasheets	1E			
	F14	Motor Datasheets				
	F15	Electrical Equipment List				
	F16	Instrument List				

VDDR			Required with Tender	Required after Award		As Built Required
			Copies	Copies	Weeks	Copies
	F17	I/O List		1E	6	
	F18	Communications Addressing and Signal Mapping List		1E	8	
G		**Inspection, Certification and MDR**				
	G01	Inspection and Test Plans	1E	1E	4	
	G02	Inspection and Test Procedures (generic)				
	G03	Quality Plans and Procedures (generic)		1E	4	
	G04	Manufacturer's Data Report (MDR) – Index / Contents		1E	4	
	G05	Manufacturer's Data Report (MDR)		1E	Delivery +2	
	G06	Material Certificates		In MDR		
	G07	Statutory Authority Certificates		In MDR		
	G08	Test Data and Other Certificates		In MDR		
	G09	Welding Procedure / Specification				
	G10	Non-Destructive Testing (NDT) Procedures				
	G11	Material Safety Data Sheets				
	G12	Commissioning Plan and Procedure		1E	Delivery - 4	
	G13	Structural Design Certification				
	G14	Performance Guarantee Statement				
H		**Shipping and Packing Details**				
	H01	Material / Shipping List		1E	Delivery -4	
	H02	Package Type and Quantity (Carton, Case, Crate)		1E	Delivery -4	
	H03	Shipping Packages – Weights and Dimensions		1E	Delivery -4	
	H04	Shipping and Handling Procedure		1E	12	
J		**Manuals**				
	J01	Installation, Operation and Maintenance Manual (Preliminary)		1E	Delivery -4	
	J02	Installation, Operation and Maintenance Manual (Final)		1E	Delivery	
K		**Spare Parts Information**				
	K01	Spare Parts Catalogue and Recommended Spares List (Copy also in Manuals)	1E	1E	8	
	K02	Recommended Spare Parts List with Pricing (Not Included in Manuals)	1E	1E	Prelim 4 Delivery +1	
	K03	Special Tools list	1E	1E	8	
M		**General**				
	M01	Control Philosophy and Logic Diagram				
	M02	Functional Description				